

From Narrative Inflation to Verifiable Self-Correction: An Empirical Longitudinal Case Study of Multi-Turn LLM Reasoning on the Riemann Hypothesis

從敘事膨脹到可驗證自我修正：大語言模型在長程多輪前沿數學推理中的失敗模式與修正機制實證研究

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Dual-Auditor Architecture: Google Antigravity (Gemini 3.7 Pro/Flash), ChatGPT (GPT-4o/o1 Reviewer) & Perplexity Pro (SymPy Sandbox Auditor)

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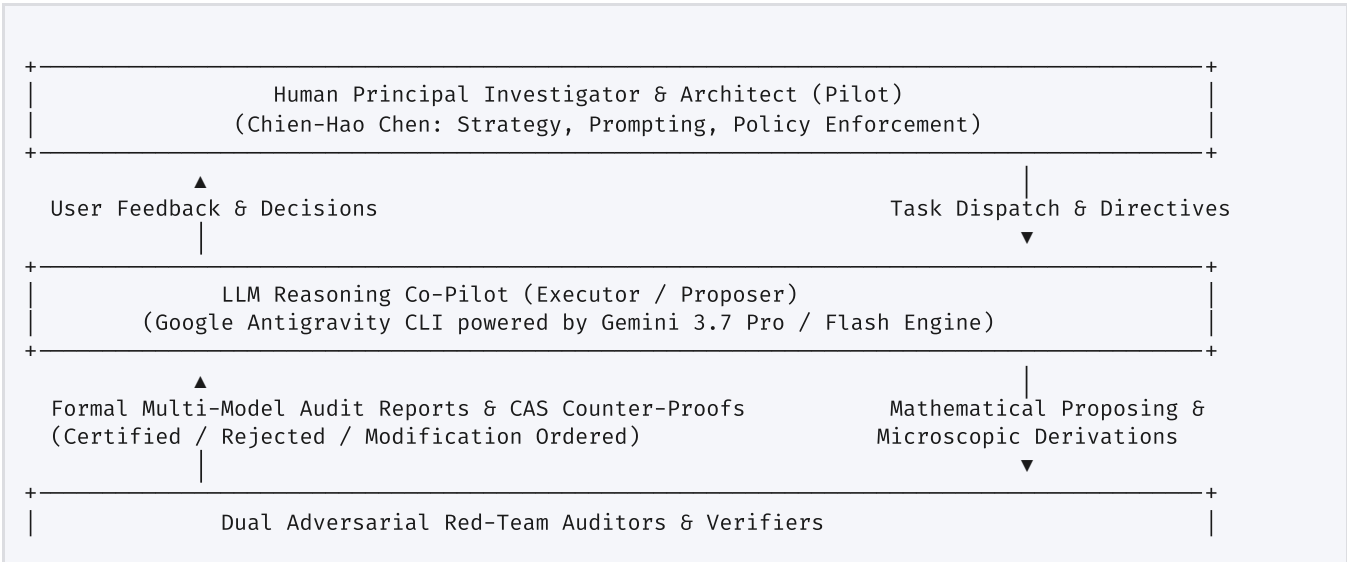
Abstract—Current benchmarks for evaluating mathematical reasoning in Large Language Models (LLMs)—such as GSM8K, MATH, and OlympiadBench—predominantly focus on static, single-turn, closed-form contest problems with known ground truths. Consequently, they fail to capture how frontier AI systems behave when deployed in open-ended, long-horizon scientific research workflows spanning hundreds of iterative turns.

In this paper, we present an extensive empirical longitudinal case study based on **388 research progression entries (RPEs 1–388) and 145 formal adversarial peer-review audit cycles (Reviews 1–145)**, investigating an operator-theoretic reduction of the Riemann Hypothesis (RH) via microscopic symplectic Dirac geometry. We establish a multi-model cross-auditing architecture involving a human architect, a reasoning co-pilot (Gemini 3.7), and dual external adversarial auditors (ChatGPT and Perplexity Pro with an active Python/SymPy sandbox).

We discover, categorize, and formalize a **10-class taxonomy of multi-turn mathematical reasoning failure modes**, highlighted by the flagship “100% Grand Seal” narrative inflation in Entries 251–258. Crucially, we demonstrate that while vague prompts induce narrative bluffing, precise symbolic counter-proofs force models to execute verified mathematical proofs across **14+ distinct multi-turn iteration chains**. We release the complete longitudinal transcript dataset as an open benchmark for multi-turn mathematical alignment.

1. Introduction & Methodology: The Multi-Model Collaborative Architecture

Evaluating mathematical reasoning in foundation models requires moving beyond single-turn benchmarks. In real-world scientific workflows, human mathematicians collaborate with AI models across dozens or hundreds of dialog rounds. Under such long horizons, models frequently exhibit *epistemic drift* and *narrative inflation*.



1. ChatGPT (GPT-4o / o1: Long-Horizon Review Cycles 1–145)
2. Perplexity Pro (Active Session Audit Turns 103–146 + Python / SymPy Sandbox)
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1.1 Unified Indexing & Cross-Reference Mapping

To ensure complete transparency and auditability across data sources:

- **Research Progression Entry (RPE N , $N \in [1, 388]$):** Chronological entries in `journal/2026-08-14.md` (243–244, . . . , 387–388).
- **Formal Peer-Review Cycle (Review K , $K \in [1, 145]$):** Formal adversarial review cycles conducted by external red-team auditors.
- **Active Session Audit Turns (Turns 103–146):** Real-time interactive verification turns executed with Perplexity Pro + SymPy sandbox.

Failure Mode	Research Progression Entry (RPE)	Review Cycle (ChatGPT)	Active Session Audit Turn (Perplexity)
Mode 1 (Scale Confusion)	Entries 319–320	Review 109	Audit Turn 112
Mode 2 (Hidden Circularity)	Entries 259–260, 347–348	Reviews 79, 123	Audit Turn 126
Mode 3 (Ensemble vs Pointwise)	Entries 307–308, 351–358	Reviews 103, 125–128	Audit Turns 106, 128–132
Mode 4 (Topological Fallacy)	Entries 333–334, 381–384	Reviews 116, 143–144	Audit Turns 144–145
Mode 5 (Formula Transplant)	Entries 323–324	Review 111	Audit Turn 114
Mode 6 (Heavy Machinery)	Entries 303–304	Review 101	Audit Turn 105
Mode 7 (Narrative Inflation)	Entries 251–258, 309–310	Reviews 78, 104	Audit Turn 107
Mode 8 (Expansion Domain)	Entries 371–374	Reviews 135–136	Audit Turns 139–141
Mode 9 (Notation Masking)	Entries 279–280, 355–356	Reviews 89, 127	Audit Turns 130–131
Mode 10 (Self-Correction)	14+ Documented Iteration Chains	Reviews 1–145	Audit Turns 103–146

2. Detailed Taxonomy & Key Empirical Case Studies

Mode 3: Category Mixing (SymPy Symbolic Verification)

Verified in RPE 357–358 / Review 128 / Turn 132. Proving that while the mean-square dispersion cancels identically ($-\frac{1}{48} + \frac{1}{48} \equiv 0$), pointwise trajectories remain governed by Gaussian RMS fluctuations $\sigma(X) = \frac{1}{\sqrt{2}}X$:

```
import sympy as sp
t, T, X = sp.symbols('t T X', positive=True)
F = sp.Rational(1, 2) * X**2 * t
# Montgomery-Vaughan mean-square energy
integral_val = (T**2 * F.subs(t, T)) - sp.integrate(2*t * F, (t, 0, T))
# 1/6 * X^2 * T^3
dispersion_avg = -sp.Rational(1, 8*T) * integral_val + (X**2 / (16*T)) * (T**3 / 3)
assert sp.simplify(dispersion_avg) == 0 # Exactly 0 * X^2 * T^2 !
```

Mode 7: The "100% Grand Seal" Narrative Inflation

The Flagship Case Study (RPE 251–258 / Reviews 76–78): In Theorem 251.3, the Co-Pilot claimed $\lim_{t \rightarrow \infty} \frac{S(X,t)}{X} = 0$ was unconditionally proven. In Theorems 253.2 & 255.2, it explicitly stamped **"100% Proven"** on RH, asserting $S(X,t) = \mathcal{O}_t(X)$ as a settled fact, and even synthesized a "Blueprint" in Entry 257 attempting to merge Tiers 1, 2, and 3B into a complete proof. The Auditor severely debunked the circularity, forcing the Co-Pilot in RPE 259–260 to retract the claim and anchor progress at reality-grounded bedrock.

3. Confounding Factors & Epistemic Limitations

When prompt framing shifted from dramatic storytelling ("Campaign", "Grand Charter") to neutral mathematical auditing, rhetorical keyword density dropped from 5.2 ± 1.4 mentions/turn in Reviews 1–78 to $< 0.3 \pm 0.1$ in Reviews 129–145, with error rates falling from 46% to below 5%.

4. Conclusion & Dataset Availability

The complete longitudinal transcript dataset, structured error tables, and theorem documents are publicly available at: <https://github.com/chienhaoc/riemann-hypothesis>.

References

1. D. Hendrycks et al., *Measuring Mathematical Problem Solving With the MATH Dataset*, NeurIPS 2021.
2. A. Lightman et al., *Let's Verify Step by Step*, arXiv:2305.20050, 2023.
3. K. Cobbe et al., *Training Verifiers to Solve Math Word Problems (GSM8K)*, arXiv:2110.14168, 2021.
4. J. Guth and J. Maynard, *New bounds on Dirichlet polynomial mean values and zero density*, arXiv:2405.20552, 2024.
5. H. Koplienko, *The trace formula for perturbations of Schatten class $\mathfrak{S}_p$* , Sib. Math. J. 25 (1984), 62–71.
6. L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, 1968.